

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech. (CE) Examination****April-May 2018****CE249 Computer Organization****Date: 09.05.2018, Wednesday****Time: 10.00 am To 01.00 pm****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1. Answer the questions below. [11]**

- A** What is operation code and operand? [1]
- B** In I/O reference instruction $p=D_7IT_3$. What is meaning of p? [1]
- C** What is binary representation of $(199)_{16}$? [1]
- D** How does microprocessor differentiate whether instruction is register, memory or I/O reference instruction? [1]
- E** State True or False: Instructions in a program are stored in non-continuous memory location. [1]
- F** Memory size is 8192×16 . What must be length of PC and AC? [1]
- G** By taking suitable 8-bit example, explain selective clear operation. [1]
- H** Write truth table for XOR gate. [1]
- I** What is immediate instructions? [1]
- J** What is use of decoder in CPU design? [1]
- K** What is direct and indirect address? How many memory accesses are required to access direct and indirect address? [1]

Q – 2. Answer any THREE. [12]

- A** Explain: Basic Computer Instruction format.
- B** Draw only: Circuit for following functionality:
Circuit which holds 4-bit of information and adds one to 4-bit at every clock pulse.
- C** What is program interrupt? How does CPU server program interrupt? What is use of flag in interrupt cycle?
- D** Draw and Explain: Common Bus using multiplexer system for basic computer.

Q – 3. Answer any TWO. [12]

- A** Draw and explain circuit for Register transfers for instruction fetch phase.
- B** Draw a circuit diagram which shows Gate structure for controlling the LD, INR and CLR ping of AC Register. (Note: Register transfer statements are given on page no:3 & 4)

C Draw a circuit for following operations:

Select			Input Y	Output $D = A + Y + C_{in}$	Microoperation
S_1	S_0	C_{in}			
0	0	0	B	$D = A + B$	Add
0	0	1	B	$D = A + B + 1$	Add with carry
0	1	0	$\overline{B}$	$D = A + \overline{B}$	Subtract with borrow
0	1	1	$\overline{B}$	$D = A + \overline{B} + 1$	Subtract
1	0	0	0	$D = A$	Transfer A
1	0	1	0	$D = A + 1$	Increment A
1	1	0	1	$D = A - 1$	Decrement A
1	1	1	1	$D = A$	Transfer A

SECTION – II

Q – 4. Answer the questions below. [11]

- A Define term: Throughput with respect to processing. [1]
- B What is auxiliary memory? [1]
- C What is locality of reference with respect to cache memory? [1]
- D Write full form of MISD. [1]
- E Where the bootstrap loader is stored? [1]
- F What is use of Match register? [1]
- G Define: Hit-Ratio [1]
- H What are the applications of vector processing? [1]
- I Write “Speedup Ratio” of a pipeline processing over an equivalent non-pipeline processing. [1]
- J State true or false: Flags are not needed for stack operations. [1]
- K List out common Page Replacement Algorithm. [1]

Q – 5. Answer any THREE. [12]

- A By taking suitable example, explain arithmetic pipeline.
- B Write the differences between RISC and CISC.
- C Draw memory and CPU connection diagram for following address map.

Component	Hexadecimal address	Address bus									
		10	9	8	7	6	5	4	3	2	1
RAM 1	0000–007F	0	0	0	x	x	x	x	x	x	x
RAM 2	0080–00FF	0	0	1	x	x	x	x	x	x	x
RAM 3	0100–017F	0	1	0	x	x	x	x	x	x	x
RAM 4	0180–01FF	0	1	1	x	x	x	x	x	x	x
ROM	0200–03FF	1	x	x	x	x	x	x	x	x	x

D What is addressing mode? Explain any four addressing modes in detail.

Q – 6. Answer any TWO.

[12]

- A** What is disadvantage of direct mapping, which is solve by set associate mapping? Explain by taking two-way set associate example.
- B** By taking suitable example, draw and explain Register based CPU organization. (Number of registers are: R1 to R7)
- C** Draw and explain flow chart (Hardware algorithm) for add and subtract operations for the sign-magnitude numbers.

Register Transfer Statements:

Fetch	$R'T_0:$	$AR \leftarrow PC$
	$R'T_1:$	$IR \leftarrow M[AR], PC \leftarrow PC + 1$
Decode	$R'T_2:$	$D_0, \dots, D_7 \leftarrow \text{Decode } IR(12 - 14),$ $AR \leftarrow IR(0 - 11), I \leftarrow IR(15)$
Indirect	$D_7IT_3:$	$AR \leftarrow M[AR]$
Interrupt:		
	$T'_0T'_1T'_2(IEN)(FGI + FGO):$	$R \leftarrow 1$
	$RT_0:$	$AR \leftarrow 0, TR \leftarrow PC$
	$RT_1:$	$M[AR] \leftarrow TR, PC \leftarrow 0$
	$RT_2:$	$PC \leftarrow PC + 1, IEN \leftarrow 0, R \leftarrow 0, SC \leftarrow 0$
Memory-reference:		
AND	$D_0T_4:$	$DR \leftarrow M[AR]$
	$D_0T_5:$	$AC \leftarrow AC \wedge DR, SC \leftarrow 0$
ADD	$D_1T_4:$	$DR \leftarrow M[AR]$
	$D_1T_5:$	$AC \leftarrow AC + DR, E \leftarrow C_{out}, SC \leftarrow 0$
LDA	$D_2T_4:$	$DR \leftarrow M[AR]$
	$D_2T_5:$	$AC \leftarrow DR, SC \leftarrow 0$
STA	$D_3T_4:$	$M[AR] \leftarrow AC, SC \leftarrow 0$
BUN	$D_4T_4:$	$PC \leftarrow AR, SC \leftarrow 0$
BSA	$D_5T_4:$	$M[AR] \leftarrow PC, AR \leftarrow AR + 1$
	$D_5T_5:$	$PC \leftarrow AR, SC \leftarrow 0$
ISZ	$D_6T_4:$	$DR \leftarrow M[AR]$
	$D_6T_5:$	$DR \leftarrow DR + 1$
	$D_6T_6:$	$M[AR] \leftarrow DR, \text{ if } (DR = 0) \text{ then } (PC \leftarrow PC + 1), SC \leftarrow 0$

	$D_7I'T_3$	$= r$ (common to all register-reference instructions)
	$IR(i) =$	$B_i (i = 0, 1, 2, \dots, 11)$
	$r:$	$SC \leftarrow 0$
CLA	$rB_{11}:$	$AC \leftarrow 0$
CLE	$rB_{10}:$	$E \leftarrow 0$
CMA	$rB_9:$	$AC \leftarrow \overline{AC}$
CME	$rB_8:$	$E \leftarrow \overline{E}$
CIR	$rB_7:$	$AC \leftarrow shr AC, AC(15) \leftarrow E, E \leftarrow AC(0)$
CIL	$rB_6:$	$AC \leftarrow shr AC, AC(0) \leftarrow E, E \leftarrow AC(15)$
INC	$rB_5:$	$AC \leftarrow AC + 1$
SPA	$rB_4:$	If $(AC(15) = 0)$ then $(PC \leftarrow PC + 1)$
SNA	$rB_3:$	If $(AC(15) = 1)$ then $(PC \leftarrow PC + 1)$
SZA	$rB_2:$	If $(AC = 0)$ then $PC \leftarrow PC + 1$
SZE	$rB_1:$	If $(E = 0)$ then $(PC \leftarrow PC + 1)$

Input-output:

	$D_7IT_3 =$	p (common to all input-output instructions)
	$IR(i) =$	$B_i (i = 6, 7, 8, 9, 10, 11)$
	$P:$	$SC \leftarrow 0$
INP	$pB_{11}:$	$AC(0 - 7) \leftarrow INPR, FGI \leftarrow 0$
OUT	$pB_{10}:$	$OUTR \leftarrow AC(0 - 7), FGO \leftarrow 0$
SKI	$pB_9:$	If $(FGI = 1)$ then $(PC \leftarrow PC + 1)$
SKO	$pB_8:$	If $(FGO = 1)$ then $(PC \leftarrow PC + 1)$
ION	$pB_7:$	$IEN \leftarrow 1$
IOF	$pB_6:$	$IEN \leftarrow 0$
